

Self-Adaptive Zwitterionic Polysilazane Coatings with Mechanical Robustness, High Transparency, and Broad-Spectrum Antiadhesion Properties

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Antiadhesive coatings have been extensively studied owing to their wide applications in biology, environment, and energy. However, developing a mechanically robust coating with broad-spectrum antiadhesion properties remains challenging. Herein, a novel strategy for preparing hard yet flexible and self-adaptive zwitterionic polysilazane coatings with broad-spectrum antiadhesion properties (anti-biofouling, anti-liquid adhesion, and anti-scaling) is proposed. The coatings are prepared by combining polysilazane with a telomer (FT) consisting of a low-surface-energy fluorine motif and hydrolysis-induced zwitterions. Before Si—OH generation in polysilazane, the fluorine motif drives the zwitterionic precursor to enrich on the surface, generating a zwitterionic layer following pre-hydrolysis. This unique design prevents the coatings from swelling in water, allowing them to adapt to diverse environments. The fluorine motif can orient toward the surface of air, providing anti-liquid adhesion capabilities, whereas the zwitterions orient underwater to endow anti-biofouling, anti-liquid adhesion, and anti-scaling capabilities. The highly cross-linked network toughened by FT contributes to the high hardness (up to 7H) and good flexibility of the coating. The chemical bonding between the coating and substrates ensures their strong adhesion (≈ 2.06 – 7.67 MPa). This study contributes to the design of mechanically robust broad-spectrum antiadhesive coatings applicable in marine industries, optical devices, pipeline transportation, and other fields.

as oil) or graffiti can considerably decrease the transparency of windows and displays, influencing their normal operation.^[4] Furthermore, the adhesion of solid substances, such as scaling in transportation pipelines or ice on outdoor facilities, severely affects the heat exchange efficiency, causing safety risks and huge economic losses.^[5–8] To address these issues, significant efforts have been devoted to the development of antiadhesive coatings. Low-surface-energy materials, such as polydimethylsiloxane (PDMS) and perfluoropolyether, have been used to endow coatings with anti-liquid or anti-solid adhesion properties.^[9–12] In addition, hydrophilic polymers with strong hydration capabilities, such as poly(ethylene glycol) (PEG) and zwitterions, serve as the foundation for various anti-biofouling coatings.^[13–20] However, they generally do not simultaneously exhibit anti-liquid, anti-solid adhesion, or anti-biofouling properties. Furthermore, the intrinsically poor mechanical properties and durability of organic polymers limit their industrial applications.

The strategy of combining organic polymers (e.g., PDMS or zwitterions)

with an inorganic network to prepare hard yet flexible coatings with antiadhesion properties as well as enhanced mechanical strength and durability has attracted significant interest.^[21–24] Polyhedral oligomeric silsesquioxanes functionalized with low-surface-tension fluorinated agents or PDMS can endow coatings with anti-liquid or anti-solid adhesion properties while maintaining their wear resistance.^[25–27] Nevertheless, such coatings lack anti-biofouling properties. Thus, PEG or zwitterionic groups are incorporated into cross-linked inorganic structures (Zr—O—Si or Si—O—Si) to endow coatings with anti-biofouling properties while maintaining high scratch resistance.^[28,29] However, the highly hydrophilic PEG or zwitterionic chains can cause the coatings to swell when exposed to water, deteriorating their mechanical properties. In addition, such hybrid coatings require complex preparation processes.^[21,25]

Polysilazane, which consists of alternating Si and N atoms, can be converted into hard coatings dominated by Si—O—Si network structures with high cross-linking densities, and the so-formed coatings have great potential as antiadhesive coatings.^[30,31] The

1. Introduction

Surface adhesion is a common phenomenon in everyday and industrial scenarios; however, undesirable adhesions (e.g., biofouling, liquids, and solids) to surfaces can result in economic losses and safety hazards.^[1] For example, biofouling on ship surfaces can increase drag during navigation, resulting in increased fuel consumption,^[2] while the biofouling of medical devices increases the risk of infection, posing a significant challenge in the medical industry.^[3] In addition, the adhesion of liquids (such

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Si–N and Si–H bonds of polysilazane coatings can interact with the hydroxyl groups on substrates, leading to excellent adhesion between the two.^[32] Furthermore, the reactive Si–N and Si–H bonds allow convenient chemical modification of polysilazane using various organic polymers, endowing the coatings with customizability. In particular, incorporating organic groups into polysilazane can improve their flexibility and enhance their antiadhesion capabilities.^[33] Therefore, coatings prepared by combining polysilazane with the aforementioned antiadhesive polymers have excellent prospects in marine industries, optical devices, pipeline transportation, and other fields, owing to their high hardness and adhesion strength.

Herein, we propose an innovative strategy for preparing a hard yet flexible and self-adaptive coating with broad-spectrum anti-adhesion properties (anti-biofouling, anti-liquid adhesion, and anti-scaling properties). The coating design involved the combination of polysilazane with a telomer (denoted as FT) consisting of a low-surface-energy fluorine motif and hydrolysis-induced zwitterions. As high-polarity zwitterions are difficult to incorporate directly into coatings because of their incompatibility with nonpolar materials and high swelling in water, we used a zwitterionic precursor, viz., tertiary carboxybetaine triisopropylsilyl ester ethyl acrylate (TCBSi), which is hydrophobic and generates zwitterions following pre-hydrolysis. TCBSi could be easily copolymerized with other nonpolar monomers and directly incorporated into coatings. FT was prepared through a free-radical reaction involving 3-mercaptopropyltriethoxysilane (KH580), dodecafluoroheptyl methacrylate (DFMA), and TCBSi. For preparing coatings, FT was mixed with polysilazane and directly applied on substrates in a humid atmosphere via hydrolysis and condensation reactions at 25 °C, because Si–N and Si–H are moisture sensitive.^[34,35] Because of its low surface energy, the fluorine motif drove the zwitterionic precursor to enrich on the coating surface, before the Si–OH bonds were formed in polysilazane. This resulted in the formation of a functional gradient coating with a strong zwitterionic hydration layer on the surface following pre-hydrolysis.^[36–38] The low zwitterion content in the coating prevented swelling and mechanical property deterioration during water exposure. In addition, zwitterions with an overall neutral charge can reduce the adsorption of scale-forming ions on the coating by decreasing their electrostatic interaction with the surface, endowing the coating with anti-scaling properties. Owing to the low surface energy of the fluorine motif and the strong affinity of the zwitterions for water, the coating exhibited self-adaptive characteristics. In the air, the fluorine motif oriented toward the surface, endowing the coating with anti-liquid adhesion capabilities, whereas the zwitterions oriented outward underwater, endowing anti-biofouling, anti-liquid adhesion, and anti-scaling capabilities. The highly cross-linked network toughened by FT contributed to the high hardness (pencil hardness of up to 7H) and good flexibility (minimum bending diameter of 1 mm) of the coatings, whereas the chemical bonding between the coating and substrates ensured their strong adhesion (≈ 2.06 – 7.67 MPa). Thus, the proposed preparation strategy provided a self-adaptive zwitterionic polysilazane coating with broad-spectrum antiadhesion and excellent mechanical properties.

2. Results and Discussion

2.1. Design and Preparation of Self-Adaptive Zwitterionic Polysilazane Coatings

Zwitterionic polymers and PEG are known to have antifouling capabilities. However, PEG is prone to oxidative degradation and its inherent amphiphilicity can influence the oil resistance of coatings.^[18] Therefore, zwitterions are being used as antifouling units in material design. However, directly introducing highly polar zwitterions into coatings is challenging because of their incompatibility and high swelling. Therefore, we developed a zwitterionic precursor, TCBSi, to solve this problem (Figure S1, Supporting Information). Subsequently, we prepared a telomer, FT, through the free-radical reaction of DFMA, KH580, and TCBSi (Figure S2, Supporting Information). Figure 1a shows the structure of polysilazane and the reaction procedure to form a highly cross-linked network. Figure 1b shows the structure of FT. The fabrication process of the self-adaptive zwitterionic polysilazane coating is illustrated in Figure 1c. The prepared FT was mixed with polysilazane in tetrahydrofuran, stirred for 1 h, and then coated on a substrate at 25 °C. Because of its low surface energy, the fluorine motif drove the zwitterionic precursor to enrich on the surface. In the presence of moisture, the Si–N bonds of polysilazane were hydrolyzed to Si–OH bonds, releasing NH_3 . Subsequently, a condensation reaction occurred between the Si–OH bonds, resulting in the Si–O–Si structure. Simultaneously, the alkoxy group of FT was hydrolyzed, and it was condensed with polysilazane (Figure S3, Supporting Information). The prepared coatings are denoted as SN-FTx (where SN represents polysilazane and x represents FT/SN-FTx wt/wt). Considering that polysilazane is an organic/inorganic hybrid material, thermogravimetric analysis was conducted to detect the content of the inorganic components (Figure S4, Supporting Information). All SN-FTx coatings yielded high residual carbon, indicating their high inorganic content. In addition, the residual carbon content decreased as the telomer (FT) content increased, indicating that the content of inorganic components decreased in the coatings. After pre-hydrolysis in water, zwitterions were generated on the coating surface. Owing to the low surface energy of the fluorine motif and the strong affinity of the zwitterions toward water, the fluorine motif would orient toward the surface in air, while the zwitterions orient underwater. Thus, the coating could be self-adaptive and exhibit anti-liquid adhesion properties in air, as well as anti-biofouling and anti-scaling capabilities underwater. Furthermore, the combination of the soft telomer and hard cross-linked networks could endow the coating with high hardness and good flexibility.

The proton nuclear magnetic resonance (^1H -NMR) spectra of TCBSi, telomer poly(KH580-co-DFMA) (F), and telomer poly(KH580-co-TCBSi) (T) are shown in Figures S5–S7 (Supporting Information) and that of FT is shown in Figure 2a. The molecular weights of the FT, F, and T telomers are listed in Table S1 (Supporting Information). The peaks at 5.83, 6.15, and 6.39 ppm in the ^1H -NMR spectrum of FT are attributed to the protons attached to the C=C bonds. Proton signals from the fluorocarbon groups were observed at 5.40 ppm, and the peaks between 2.3 and 2.8 ppm were assigned to the protons of $-\text{CH}_2-$

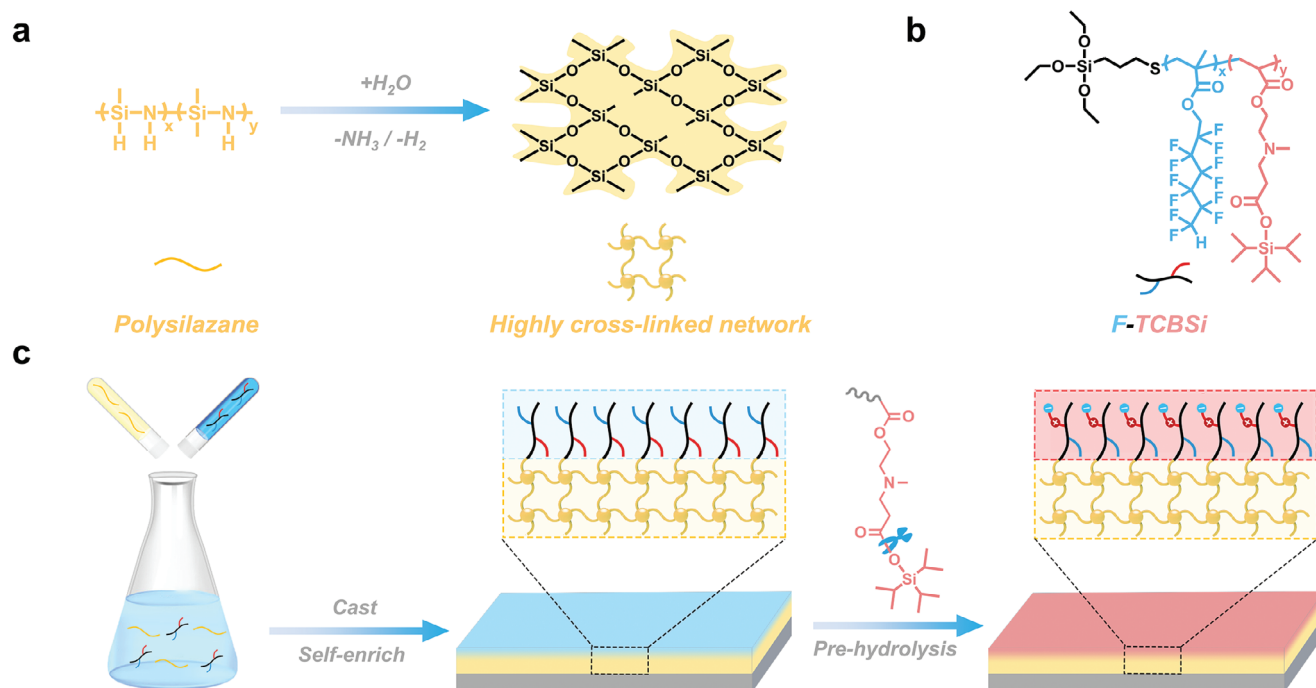


Figure 1. Design strategy and preparation of the coatings. a) Structure of polysilazane and the reaction procedure to form a highly cross-linked network. b) Structure of the telomer FT. c) Schematic of the preparation of self-adaptive zwitterionic polysilazane coatings.

and $-CH_3$ linked to the nitrogen atom. The decrease in the intensity of the $C=C$ proton peaks and the appearance of the peaks corresponding to the fluorinated and alkyl groups connected to the nitrogen atom indicated the successful synthesis of FT. When FT was mixed with polysilazane at 25 °C and coated on glass, a highly transparent coating was obtained (transmittance > 90%) (Figure 2b).

Figure 2c and Figure S8 (Supporting Information) show the surface and cross-sectional scanning electron microscopy (SEM) images of the SN-FTx coatings; all the coatings presented a flat surface and had a thickness of ≈ 50 μm . X-ray photoelectron spectroscopy (XPS) was conducted to examine the surface composition of the SN-FTx coatings (Figure 2d). The XPS peak at 686 eV is attributed to F 1s of the fluorocarbon groups. The F 1s peak intensity increased with increasing FT content in the SN-FTx coatings. More importantly, all SN-FTx coatings exhibited a higher F content on the surface than in the bulk (Table S2, Supporting Information), indicating that the fluorocarbon groups were self-enriched on the surface during coating formation because of their low surface energies. Figure 2e shows the high-resolution N 1s spectrum of the SN-FT30 coating before and after hydrolysis. Compared with that before hydrolysis, new peaks at 399.5 eV (attributed to N^+) appeared, indicating the hydrolysis of TCBSi. Figure 2f presents the attenuated total reflection infrared (ATR-IR) spectra of the SN-FT30 coating before and after hydrolysis. The appearance of the carboxyl (≈ 1560 cm^{-1}) and hydroxyl (≈ 3410 cm^{-1}) vibrational peaks after hydrolysis further proved the formation of zwitterions on the surface.^[39,40]

Figure 2g,h shows the advancing, static, and receding water contact angles (WCAs) of the SN-FTx coatings. Before hydrolysis, the static WCA of SN-0 (pristine polysilazane coating) was

$\approx 105^\circ$, indicating that the coating was hydrophobic. After 30 min of soaking in water, the static WCAs of all the coatings, with the exception of SN-0 and SN-F30 (Figure S9, Supporting Information), decreased considerably. Note that a typical zwitterionic surface is expected to have a WCA of less than 20° , but the WCA of the coating subjected to hydrolysis is 70° – 80° for the best coating. This is probably because the Si–O–Si network and F-containing motifs also affect the surface wettability. In fact, the WCA is a macroscopic measure of the hydrophobicity or hydrophilicity of a coating, and the advancing WCA reflects the hydrophobicity of coatings, whereas the receding WCA reflects the hydrophilicity.^[41,42] Compared with the static WCAs, the advancing WCAs of the coatings increased significantly before hydrolysis, whereas the receding WCAs of the coatings decreased significantly after hydrolysis, as the telomer (FT) content increased. The receding WCAs of our coatings were $\approx 52^\circ$ after hydrolysis, indicating the formation of a relatively hydrophilic surface. This further demonstrates that zwitterions were generated on the surface. The enrichment of zwitterionic precursors on the surface and subsequent pre-hydrolysis can prevent the zwitterions from being trapped within the coating and mitigate the adverse effects of excessive zwitterions on the mechanical performance of the coating.

2.2. Mechanical Properties

The durability of coatings used in marine industries, optical devices, pipeline transportation, and other fields is determined based on their mechanical properties. Therefore, we initially investigated the mechanical properties of the coatings formed

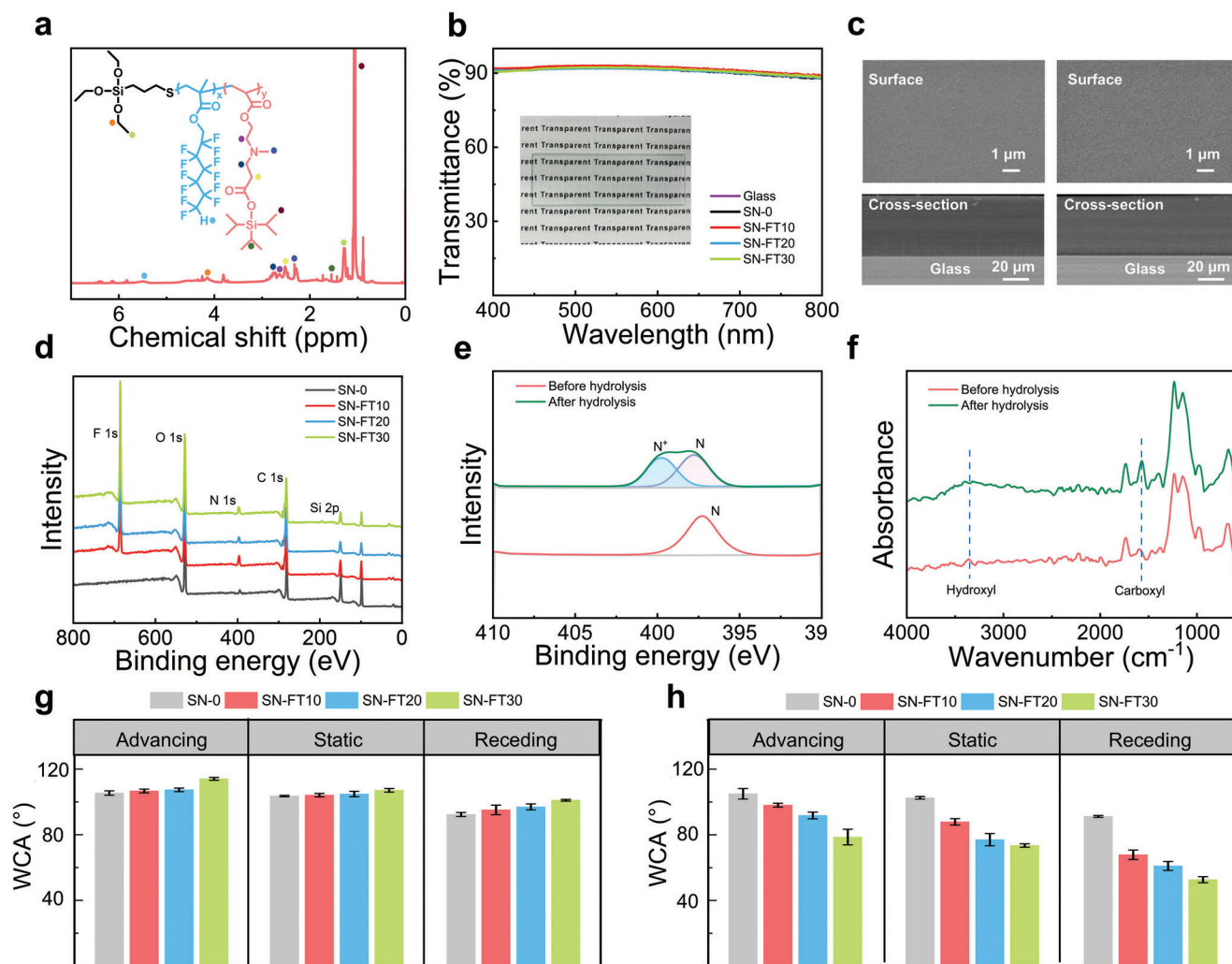


Figure 2. Characterization of self-enrichment and hydrolysis-induced zwitterions. a) ^1H -NMR spectrum of FT with assigned protons. b) Transmittance spectra of the SN-FTx coated glass substrates (inset photograph shows the transparent SN-FT30 coated glass). c) High-resolution SEM images of SN-0 (left) and SN-FT30 (right) coatings. d) XPS wide-scan spectra of the SN-FTx coatings. e) N 1s high-resolution spectra of the SN-FT30 coating before and after hydrolysis. f) ATR-IR spectra of the SN-FT30 coating before and after hydrolysis. g) WCAs of the SN-FTx coatings before and h) after hydrolysis.

on glass substrates using pencil hardness tests. As shown in **Figure 3a**, the pencil hardness values for SN-0, SN-FT10, SN-FT20, and SN-FT30 were 8, 9, 9, and 7 H, respectively. Then, the quantitative hardness (H) and elastic modulus (E^*) of the coatings on glass substrates were measured using a nanoindenter. In the evaluation, the nanoindentation tip was inserted to a depth of 500 nm into each coating, and the harder the coating was, the greater the loading force required to damage its surface. **Figure 3b** shows that SN-0 had a hardness of 248.73 MPa and a modulus of 2.24 GPa. As the FT content increased, the hardness of the SN-FTx coatings initially increased and then decreased; SN-FT10, SN-FT20, and SN-FT30 exhibited hardness values of 425.05, 340.85, and 223.05 MPa, respectively, and moduli of 3.04, 3.16, and 2.13 GPa, respectively. Both the hardness and modulus showed a similar trend of variation with the FT content. This is because the alkoxy group of FT served as a curing agent, increasing the cross-linking density of polysilazane. However, the

gradual increase in the FT content, which served as a plasticizer, caused the hardness to decrease.

The wear resistances of the coatings were determined using a wear tester. **Figure 3c** shows that the SN-FTx coatings with higher surface hardness had a lower abrasion value (≈ 30 mg) than the PDMS coating (507.4 mg). Because polysilazane had a higher crosslinking density, which increased its stiffness, the SN-FTx coatings exhibited higher resistance to abrasion than the PDMS coating. However, as the surface FT content increased, the abrasion value of the SN-FTx coatings increased gradually (26.5, 29.6, 36.1, and 41.5 mg for SN-0, SN-FT10, SN-FT20, and SN-FT30, respectively). Nevertheless, SN-0, SN-FT10, SN-FT20, and SN-FT30 exhibited sufficient wear resistance to meet the general requirements. For a coating, its hardness and flexibility are mutually exclusive. Inorganic coatings (e.g., ceramics) have high hardness but low flexibility, whereas organic coatings (e.g., polymers) have good flexibility but low hardness. The

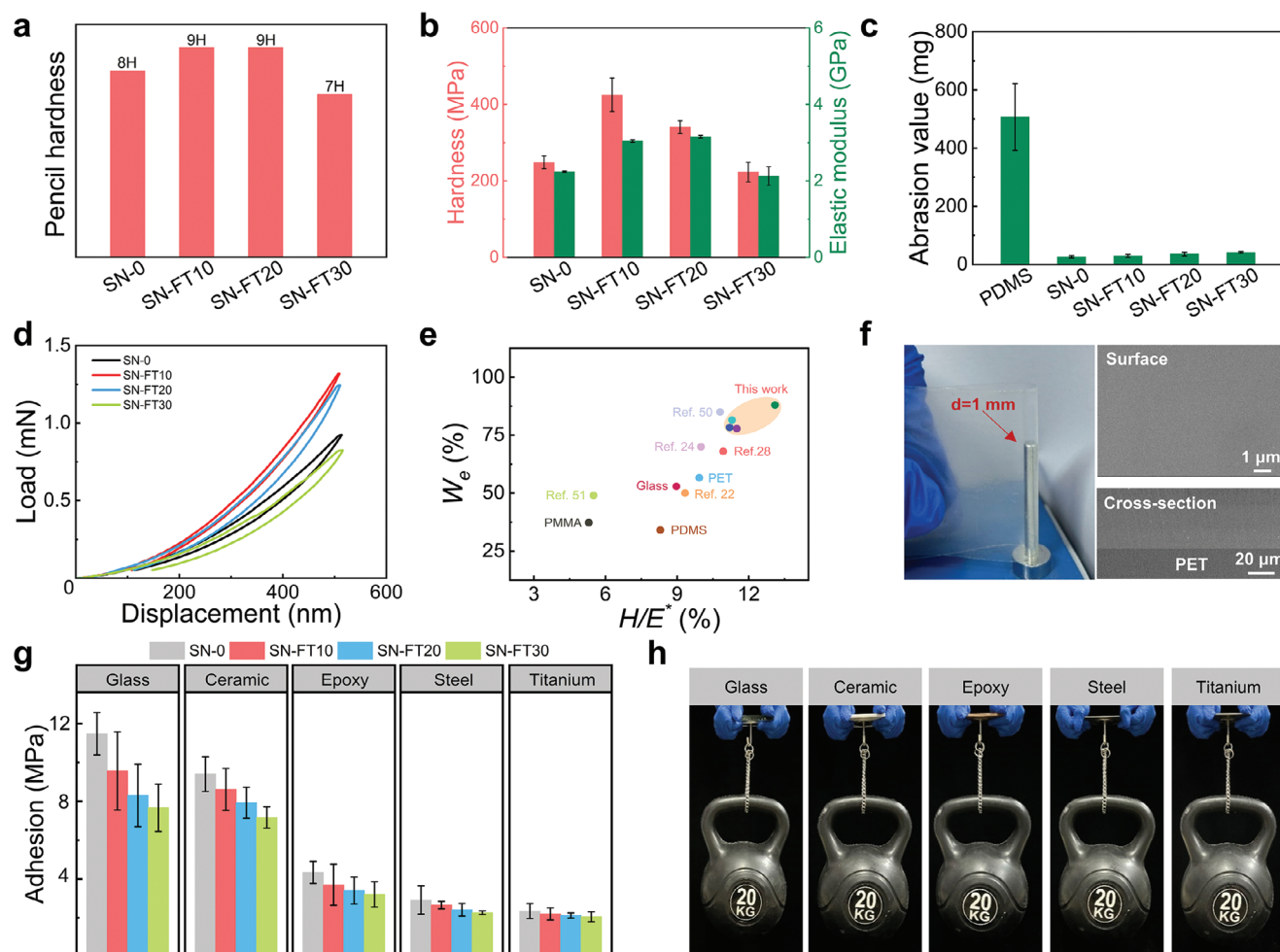


Figure 3. Mechanical properties of the SN-FTx coatings. a) Pencil hardness, b) hardness and elastic modulus, c) abrasion value, and d) load-displacement curves of the SN-FTx coatings. e) Calculated H/E^* and W_e of various samples.^[22,24,28,50,51] f) After bending the SN-FT30 coated PET film, the coating still firmly adhered to the film with no cracks. g) Adhesion strengths of the SN-FTx coatings to glass, ceramic, epoxy, steel, and titanium substrates. h) Photographs of the SN-FT30 coating on different substrates withstanding a 20 kg kettlebell.

simultaneous achievement of ceramic-like hardness and polymer-like flexibility in a coating is challenging. Generally, the materials with a low value of E^* satisfying hardness/modulus (H/E^*) ≥ 0.1 and bending recovery rate (W_e) $\geq 60\%$ can be considered as flexible hard coatings.^[21,43,44] Figure 3d shows the load-displacement curves determined using the nanoindenter; the corresponding H/E^* and W_e values are listed in Table S3 (Supporting Information). Clearly, all coatings showed $H/E^* \geq 0.1$ and $W_e \geq 60\%$, fulfilling the essential requirements of hard yet flexible coatings (Figure 3e).^[45–47] Considering that nano-indentation measurements often involve systematic errors, we used some typical standard controls such as a PDMS coating, glass, polymethyl methacrylate (PMMA), and polyethylene terephthalate (PET) to verify the accuracy of the measurements (Table S4, Supporting Information). The measured H and E^* values of these materials are close to those reported previously, validating that the nano-indentation results in this study are reasonable.^[44,48,49] This fact was supported by visual experiments of the SN-FTx coatings, where these high-hardness (>7H) coatings could be bent (diameter = 1 mm, 100 cycles) with no cracks being observed on

their surfaces during folding and after recovery, which confirmed that the coatings have high hardness as well as good flexibility (Figure 3f; Figure S10, Supporting Information).^[22,24,28,50,51]

Adhesion strength is an important parameter for coatings. The adhesion strengths of the SN-FTx coatings on various substrates, such as glass, ceramic, epoxy, steel, and titanium, were measured according to ASTM D4541-09 (Figure 3g). SN-0 exhibited the highest adhesion strength to glass (≈ 11.48 MPa), followed by ceramic (≈ 9.40 MPa), epoxy (≈ 4.34 MPa), steel (≈ 2.92 MPa), and titanium (≈ 2.35 MPa). This result indicates that polysilazane exhibited high adhesion performance, which is attributed to the chemical interactions of the Si–N and Si–H bonds with the hydroxyl groups on the substrates. For glass and ceramics, they have more hydroxyl groups, which leads to more chemical bonds between the coating and substrates for better adhesion. As the FT content increased, the adhesion strength gradually decreased because some of the telomers were trapped at the bottom of the coating during film formation, thereby influencing the adhesion between polysilazane and substrates. The SN-FT30 coating exhibited good adhesion to glass (≈ 7.67 MPa), ceramic (≈ 7.17

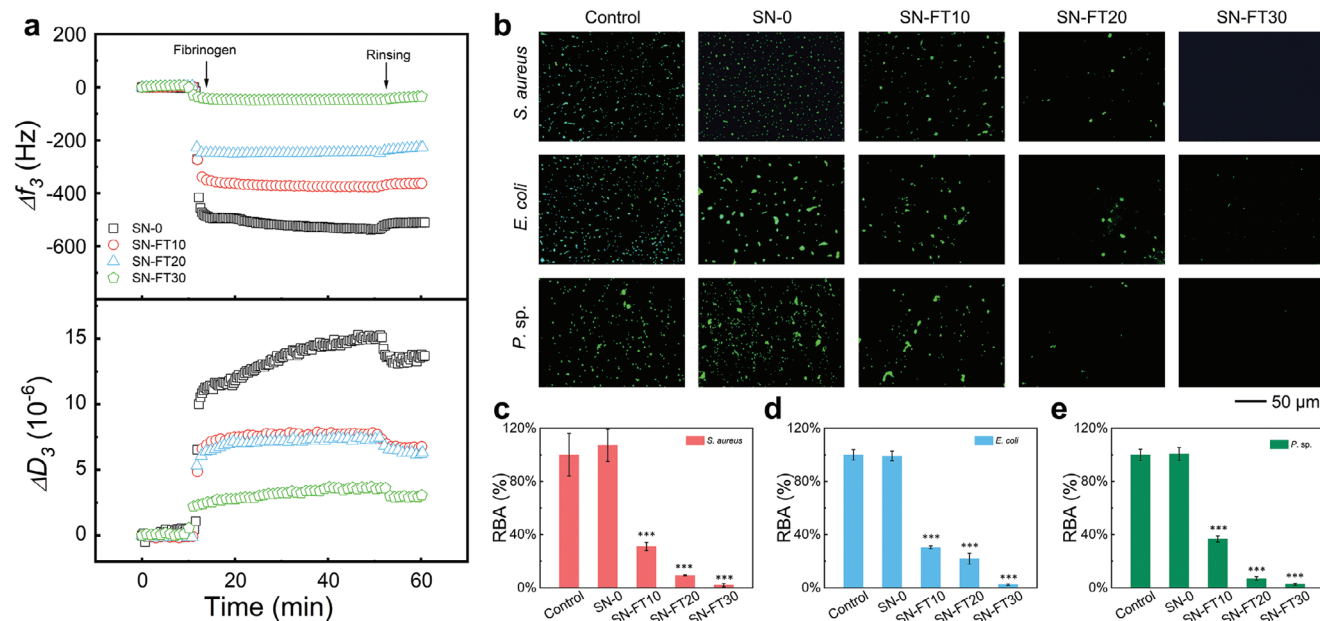


Figure 4. Antifouling properties of the SN-FTx coatings. a) Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on the SN-FTx coatings. b) Fluorescence microscopy images of *S. aureus*, *E. coli*, and *P. sp.* settled on the SN-FTx coated surfaces and the relative bacteria adhesion (RBA) of c) *S. aureus*, d) *E. coli*, and e) *P. sp.* The data are presented as mean values $\pm$ SD, $n = 3$, $***p < 0.001$.

MPa), epoxy (≈ 3.21 MPa), steel (≈ 2.26 MPa), and titanium (≈ 2.06 MPa). Figure 3b shows that the substrates coated with SN-FT30 could easily withstand a kettlebell weighing 20 kg. Note that the adhesion strength evaluated with 20 kg weights is only for the visual presentation of strong adhesion, and the results are not quantitative.

2.3. Anti-Biofouling Properties

The initial process of biofouling involves protein adsorption. As some fouling organics secrete protein-based adhesives for the colonization of submerged surfaces, protein resistance is essential for antifouling coatings.^[52] Thus, the protein resistance of the SN-FTx coatings was investigated in an artificial seawater (ASW) environment using a quartz crystal microbalance (QCM-D). Fibrinogen, a high-molecular-weight hydrophobic protein that strongly adsorbs to hydrophobic surfaces, was used as the model protein. Figure 4a shows the time dependence of the frequency shift (Δf) and dissipation shift (ΔD) evaluated using the QCM-D for fibrinogen adsorption on SN-FTx coatings. For SN-0, a notable decrease in Δf and an increase in ΔD were observed with the introduction of fibrinogen solution (1.0 mg mL^{-1} ASW), indicating substantial fibrinogen adsorption on the surface. As the FT content in the coating increased, the variations in the Δf and ΔD decreased. The negligible changes in the Δf and ΔD values of the SN-FT30 coating indicate its superior protein resistance, which is attributed to the strong hydration of the zwitterions on the coating surface. Figure S11 (Supporting Information) shows that SN-FT30 coating (polysilazane containing 30 wt.% F telomer) exhibited a lower level of decrement in Δf and increment in ΔD than SN-0. This is probably because the low-surface-energy fluorine motif has a fouling release capacity. The SN-T30 coating

(polysilazane containing 30 wt.% T telomer), which had the highest zwitterionic precursor content, exhibited lower protein resistance than SN-FT30. This can be explained by the limited surface hydration of the SN-T30 coating, with some zwitterionic precursors confined in the cross-linked network rather than being exposed on the coating surface. Therefore, the incorporation of FT into polysilazane endowed the coatings with anti-protein properties. Furthermore, the low-content zwitterions imparted the coating with robust hydration capabilities via the self-enrichment effect of the fluorine motif. Our coating design effectively mitigated the adverse effects of excessive zwitterions on the mechanical performance of the coatings, thereby ensuring their durability and stability.

Figure 4b shows the fluorescence microscopy images of the SN-FTx coatings after immersion in suspensions of *Staphylococcus aureus* (*S. aureus*), *Escherichia coli* (*E. coli*), and marine bacteria *Pseudomonas sp.* (*P. sp.*), followed by relative bacterial adsorption (RBA) on the surfaces (Figure 4c,d,e, respectively). For SN-0, numerous bacteria were observed on the surface, indicating that polysilazane was susceptible to bacterial adhesion. As the FT content in the coating increased, the RBA gradually decreased because more zwitterionic precursors were enriched on the surface, forming a hydration layer via hydrolysis-induced zwitterions, which prevented bacterial adhesion. Compared with SN-0, almost no bacteria adhered to the SN-FT30 coating, indicating that a strong zwitterionic hydration layer was formed on the coating surface, which effectively resisted bacterial adhesion. To exclude the effect of coating toxicity on the anti-biofouling, we used bacterial log reduction tests to evaluate the toxicity of the coatings as described previously.^[38,53] As shown in Figure S12 (Supporting Information), the colony counts of *S. aureus*, *E. coli*, and *P. sp.* of SN-FT30 were similar to those of the control samples, indicating that the coatings exhibit no bactericidal activity or toxicity. In

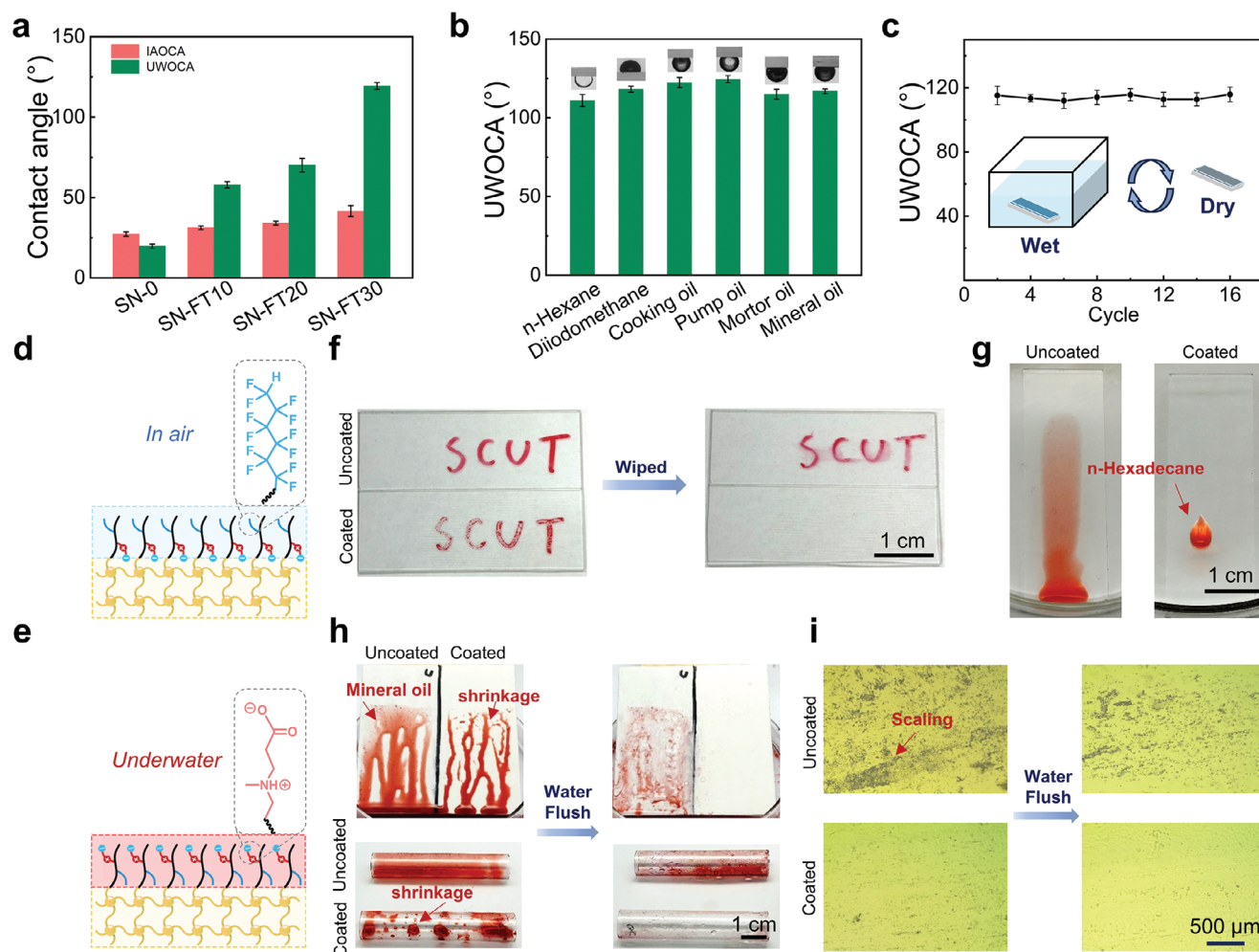


Figure 5. Anti-liquid adhesion and anti-scaling properties of the SN-FTx coatings. a) IAOCAs and UWOCAs of the SN-FTx coatings after hydrolysis. b) UWOCAs of the SN-FT30 coating. c) UWOCAs of the SN-FT30 coating after drying–rehydration cycles. d) Schematic of the fluorine motif orients toward the surface of air. e) Schematic of the zwitterions orient toward the surface underwater. f) Anti-smudge capability of the SN-FT30 coating. g) Photographs illustrating the resistance of the SN-FT30 coating to n-hexadecane in air. h) Anti-liquid adhesion of the SN-FT30 coating before and after water flush. i) Anti-scaling properties of the SN-FT30 coating before and after water flush.

other words, the low bacterial count is a result of anti-adhesion, not the toxicity of the coating. As discussed above, the surfaces of the coatings developed in this study are not as hydrophilic as a typical zwitterionic surface. The excellent anti-biofouling performance of our coating indicates that the low amount of zwitterionic moieties at the surface is sufficient to form a hydration layer that resists protein fouling and bacterial adhesion.^[54] Previous studies have also shown that a relatively less hydrophilic surface (WCA of 70°–90°) can resist protein fouling and bacterial adhesion.^[55,56]

2.4. Anti-Liquid Adhesion and Anti-Scaling Properties

Broad-spectrum antiadhesion properties are important for coatings and essential for their application in biology, environment, and energy. Therefore, we investigated the anti-liquid adhesion and anti-scaling properties of the coatings. The in-air oil contact

angle (IAOCA) and underwater oil contact angle (UWOCA) of the SN-FTx coatings after hydrolysis were determined using n-hexadecane as the model oil (Figure 5a; Figure S12, Supporting Information). As the FT content of the coating increased, the IAOCA increased, indicating that more fluorine motifs existed on the coating surface in the air. When the coatings were submerged in water, the UWUCA increased with increasing FT content, indicating that more zwitterions were present on the coating surface underwater. Further, the UWOCAs of the coatings were measured using various oils, including n-hexane, diiodomethane, cooking oil, pump oil, motor oil, and mineral oil (Figure 5b). All oils formed spherical shapes on the coating surfaces underwater, with the UWOCAs measuring ≈120°. The UWOCAs of typical diiodomethane were maintained at ≈120° after 16 drying–rehydration cycles, indicating the stable underwater oleophobicity of the coatings (Figure 5c). As discussed above, the coatings were prepared by combining polysilazane with FT consisting of a low-surface-energy fluorine motif and

hydrolysis-induced zwitterions. During the formation of the cross-linked films, the low-surface-energy fluorine motif drove the zwitterionic precursor to enrich on the surface, yielding zwitterionic chains following pre-hydrolysis. In the air, the low surface energy of fluorine plays a dominant role in chain orientation, driving the fluorine-rich chains to orient toward the surface (Figure 5d). In water, the highly hydrophilic zwitterionic chains interact strongly with water, playing a dominant role in chain orientation, thus driving the zwitterionic chains to reorient toward the surface (Figure 5e). Direct evidence of these changes was obtained using ATR-IR spectroscopy to evaluate the changes in the carboxyl and C—F bond vibrational peaks of the SN-FT30 coating in wet and dry conditions for multiple samples (Figure S13, Supporting Information). To achieve the wet condition, the ATR-IR spectrum was recorded immediately after the sample was taken out of the water. The typical carboxyl ($\approx 1600\text{ cm}^{-1}$) and C—F bond peaks ($\approx 680\text{ cm}^{-1}$, wagging vibrations) could be clearly observed both in wet and dry conditions because the ATR-IR technique provides information on a micrometer-thick scale for coatings.^[57] Note that XPS is a more sensitive tool for obtaining surface information, but requires a high level of vacuum, which is not suitable for examining surfaces in wet conditions.^[58] Anyhow, as the condition was changed from wet to dry, the intensity of the carboxyl peaks decreased while that of the C—F peaks increased. The average ratio of the integrated peak intensities of the C—F bond to the carboxyl group across multiple samples with the standard division increased from 2.16 ± 0.06 to 4.47 ± 0.25 (Table S5, Supporting Information). Moreover, the peaks between 1140 and 1235 cm^{-1} attributed to the overlap of C—O—C and C—F (stretching vibrations) also became intense from wet to dry conditions.^[59–61] These results clearly support our claim that the mobility of the F-containing and zwitterionic motifs allows them to move inward and outward according to the environment they are exposed to (air vs water). In other words, the coatings could exhibit self-adaptive properties, resulting in anti-liquid adhesion properties in air and underwater.

As shown in Figure 5f, inscriptions were made using an oily pen on uncoated glass and SN-FT30-coated glass. The writing on the uncoated glass was clear and difficult to erase using a napkin as the oil had completely spread. In contrast, the writing on the coated glass could be easily wiped off, because the oil aggregated into small droplets. This indicates that the fluorine motif endowed the coating with anti-smudge properties, protecting the surface from ink contamination in the air. The anti-liquid adhesion properties in air were further studied using n-hexadecane as the model oil. When n-hexadecane was dropped onto uncoated glass, the oil adhered and spread on the surface. In contrast, the oil droplet on the SN-FT30-coated glass rolled off smoothly, leaving no residue (Figure 5g). Figure 5h shows the anti-liquid adhesion of the coatings from air to water. When the mineral oil was poured onto SN-FT30-coated ceramic and PMMA tubes, the oil aggregated. Meanwhile, the oil was dispersed on the uncoated substrates. Upon flushing with water, the oil on the coatings was easily washed off, indicating that the zwitterions on the surfaces endowed the coatings with excellent underwater anti-liquid adhesion properties.

Considering that the hydration layer and overall neutral charge of the zwitterionic polymers can reduce the adsorption of scale-forming ions on the coatings by decreasing their electrostatic at-

traction, the coatings were expected to have anti-scaling properties as well. To evaluate the anti-scaling properties of the SN-FT30 coating, the samples were immersed in a CaSO_4 solution for 24 h at 65°C . The optical images shown in Figure 5i indicate a clear difference between the coated and uncoated glass substrates; the CaSO_4 -scale build-up on the uncoated substrate was considerably more. In addition, the flat surface of the coating indicated that it did not swell even in hot water, implying that the low zwitterion content in the coating prevented its swelling. After being flushed with water, minimal CaSO_4 remained on the coating, because the low-surface-energy fluorine facilitated salt removal after scaling. Overall, FT endowed the coatings with anti-liquid adhesion and anti-scaling properties, thereby demonstrating their application potential in pipeline transportation.

3. Conclusion

We developed hard yet flexible and self-adaptive zwitterionic polysilazane coatings by combining polysilazane with a telomer comprising a low-surface-energy fluorine motif and hydrolysis-induced zwitterions. The prepared coatings are transparent, mechanically robust, and have broad-spectrum antiadhesion properties (anti-biofouling, anti-liquid adhesion, and anti-scaling). The high crosslinking density endowed the coatings with high hardness, and the Si—N and Si—H groups of polysilazane facilitated strong adhesion to the substrates. The good flexibility of the coatings was attributed to the methyl groups of the polysilazane and the soft telomer. Owing to the low surface energy of the fluorine motif and the strong affinity of the zwitterions to water, the coatings exhibited self-adaptive capability. The fluorine motif is oriented toward the surface, endowing anti-liquid adhesion capabilities to the coatings in air, whereas the zwitterions oriented underwater endow anti-biofouling, anti-liquid adhesion, and anti-scaling capabilities. This study offers a new perspective on the design and fabrication of mechanically robust and broad-spectrum antiadhesive coatings that have potential applications in marine industries, optical devices, pipeline transportation, and other fields.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

antiadhesion, hard yet flexible, polysilazane, self-adaptive, zwitterions

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